Answers and talk notes

Determine Equivalent Ratios Using Graphs

What your student is learning

Tonight we read a ratio graph and say out loud what each point MEANS — especially the point (1, r) and the origin (0, 0).

Answers

Ask your student to explain before you look.

1. **C.** 7. The point (1, 7) means 1 minute goes with 7 pages, so 7 pages in 1 minute.
2. **B.** 5 bags hold 30 marbles. In (5, 30) x = 5 bags and y = 30 marbles, so 5 bags hold 30 marbles.
3. **D.** (1, 9). In 1 hour you go 9 miles, so the unit rate point is (1, 9).
4. **A.** (4, 24). y is 6 times x, so for x = 4 the y-value is 4 × 6 = 24, the point (4, 24).
5. **C.** 0 hours go with 0 miles. 0 hours go with 0 miles: in no time you travel no distance.
6. **B.** 40. The unit rate is 5, so y = 8 × 5 = 40.

Talk about it

What does the point with x = 1 tell you about this rate?

Home mission

Pretend each bag holds 6 apples. Ask your student to say the point for 4 bags as a sentence. Then ask: what does the point (0, 0) mean?

When they are stuck, ask

- Which quantity is on the x-axis? Read that number first.
- What does the point where x is 1 tell us?
- Can you say this point as a sentence about real things?

Try not to say

- The order of the two numbers does not really matter.
- The origin is just where the graph starts, it has no meaning.
- Read the bigger number first, it looks neater.